4. Intersection of Two Sorted Arrays · O(m + n) Time, O(1) Space

```
#include <stdio.h>

int main() {

int T;

scanf("%d", &T);

while (T--) {

int n1, n2;

scanf("%d", &n1);

int a[n1];

for (int i = 0; i < n1; i++)

scanf("%d", &a[i]);

scanf("%d", &n2);

int b[n2];

for (int i = 0; i < n2; i++)

scanf("%d", &b[i]);

int i = 0, j = 0;

int first = 1;

while (i < n1 && j < n2) {

if (a[i] == b[j]) {
if (!first)
```

```
printf(" ");

printf("%d",

a[i]); first = 0;

i++;

j++;

}

else if (a[i] < b[j]) {

i++;

}

else {

j++;

}

}

printf("\n");

}

return 0;

}
```

5. Pair with a Given Difference · O(n²) Time, O(1) Space

```
#include <stdio.h>
```

```
int main() {

int n, k;

scanf("%d", &n);

int a[n];

for (int i = 0; i < n; i++)

scanf("%d", &a[i]);

scanf("%d", &k);

int i = 0, j = 1;

while (i < n && j < n) {

if (i == j) {

j++;

}

else if (a[j] - a[i] == k) {

printf("1");

return 0;

}

else if (a[j] - a[i] < k) {

j++;

}
```

```
else {
i++;
}
}
printf("0");
return 0;
}
```

6. Pair with a Given Difference · O(n) Time, O(1) Space

```
#include <stdio.h>
int main() {
int n, k;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++) {
scanf("%d", &a[i]);
}
scanf("%d", &k);
int i = 0;
int j = 1;
```

```
while (i < n && j < n) {
if (i == j) {
j++;
continue;
}
int diff = a[j] - a[i];
if (diff == k) {
printf("1");
return 0;
}
if (diff < k)
j++;
else
i++;
}
printf("0");
return 0;
}
```